

KOHLER × MIT WORLD PEACE UNIVERSITY

AI Research Lab Program · Track 1

KOHLER AI Bathroom Designer & Planner

An AI-assisted constrained-recommendation system for complete bathroom configuration — from customer constraints to a feasible, budget-aware product bundle.

AI Understanding + Verified Retrieval + Optimization

The Problem & The Objective

The Problem

Bathroom product selection is fragmented. A customer must understand dimensions and layout, identify what physically fits, compare specifications, match aesthetics, stay within budget, and check that chosen products are compatible with one another — a multi-constraint decision problem. Traditional product search treats products individually rather than solving the complete bathroom configuration as a system.

The Objective

Build an interactive AI design assistant that takes a customer's bathroom dimensions, layout/image, budget, design theme and product requirements, and automatically generates personalized, feasible, budget-aware KOHLER product bundle recommendations spanning faucets, smart toilets, thermostatic showers and vanities.

The variables that must be reasoned about together:

Dimensions &
Layout

Product
Compatibility

Budget

Design
Aesthetics

Functional
Requirements

Catalogue
Availability

In short — the proposed system converts a fragmented, multi-constraint search process into a single AI-assisted constrained recommendation problem.

Proposed Solution

An AI-powered bathroom-planning pipeline — not a product generator, a feasibility-optimizer.

1

User Constraint Collection

Dimensions, budget, theme & requirements from the customer

2

Bathroom Image / Layout Analysis

Spatial information extracted from an uploaded photo or layout

3

Product Catalogue Retrieval

Candidate KOHLER products retrieved from the vector store

4

AI-Based Reasoning

Requirements and space understood, constraints structured

5

Compatibility Checking

Candidates filtered for physical & specification fit

6

Constraint-Based Optimization

Budget, space & preference constraints applied together

7

Product Bundle Generation

A feasible, complete combination is assembled

8

Explanation of Recommendation

Reasoning returned alongside the recommended bundle

Not aesthetics alone — the system generates a feasible and optimized product combination based on the customer's actual constraints — not simply attractive products.

Key Innovation — Separating Reasoning from Decisions

AI Handles

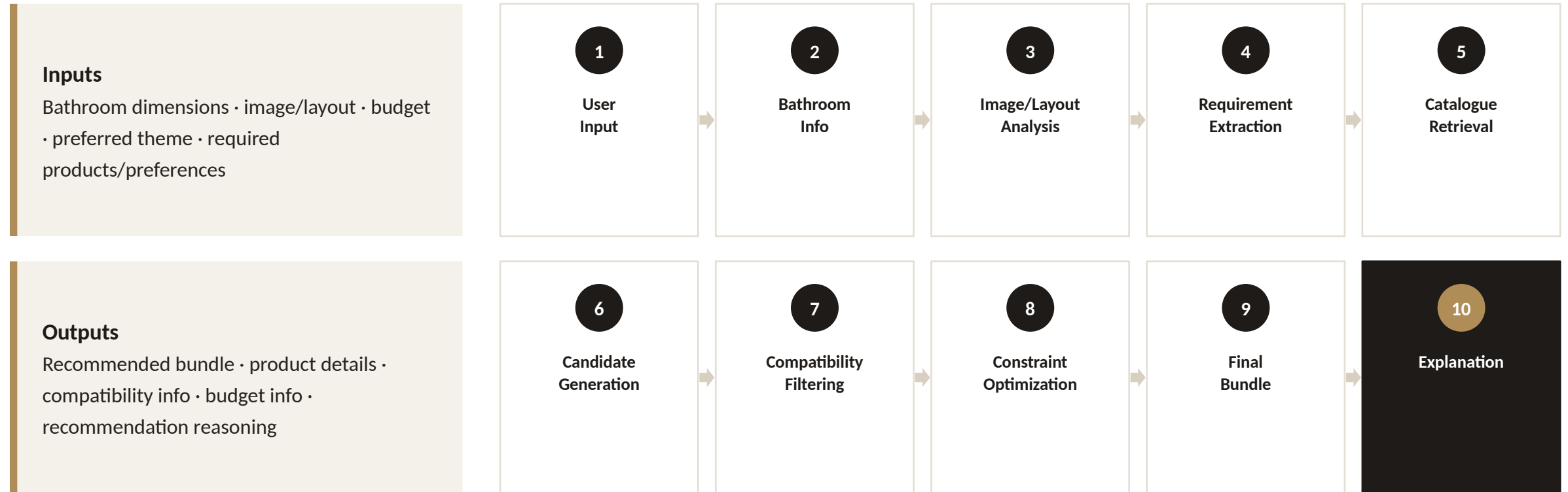
- Understanding user requirements
- Interpreting bathroom images/layouts
- Understanding natural-language preferences
- Retrieving and reasoning over relevant product information
- Generating explanations

Deterministic / Optimization Handles

- Budget constraints
- Product compatibility
- Dimension constraints
- Bundle scoring
- Constraint satisfaction
- Final recommendation selection

Why it matters — *this reduces the risk of relying solely on generative AI for product-selection decisions.*

User Workflow



End to end — ten steps carry the customer from raw inputs to an explained, feasible product bundle.

System Architecture

1	Frontend	Collects dimensions, images/layouts, budget & preferences; displays recommended bundles
2	Backend	Processes inputs, manages the recommendation workflow, connects AI services, calls retrieval & optimization, returns structured recommendations
3	Image Processing Layer	Bathroom image → image understanding → layout/objects/dimensions extraction → structured bathroom constraints
4	AI Layer — Bedrock + Nova (integration-ready)	Natural-language understanding · image understanding · requirement extraction · reasoning · recommendation explanation
5	Retrieval Layer — ChromaDB (integration-ready)	Catalogue transformed into searchable representations — category, specs, dimensions, features, style, compatibility — grounded in verified product data
6	Optimization Layer	Evaluates candidates against budget, dimensions, compatibility, preferences, category requirements & theme consistency to produce the final feasible bundle

AI & Technology Stack

Frontend

React 19 • Vite 7 • TypeScript • Tailwind CSS • shadcn/ui — the bathroom-planning UI

Backend

Node.js • Express • tRPC • TypeScript • Zod • Drizzle ORM — type-safe API & orchestration

Data

MySQL / TiDB • S3-compatible storage • KOHLER catalogue — application & media persistence

AI & Intelligence

Amazon Bedrock / Nova (integration-ready) • Python / FastAPI • ChromaDB (integration-ready) • RAG

Core Engines

Catalogue retrieval • spatial intelligence • compatibility engine • colour intelligence • lookbook intelligence • constraint & budget optimization

DevOps & Testing

Docker • Vitest • Pytest • Git/GitHub

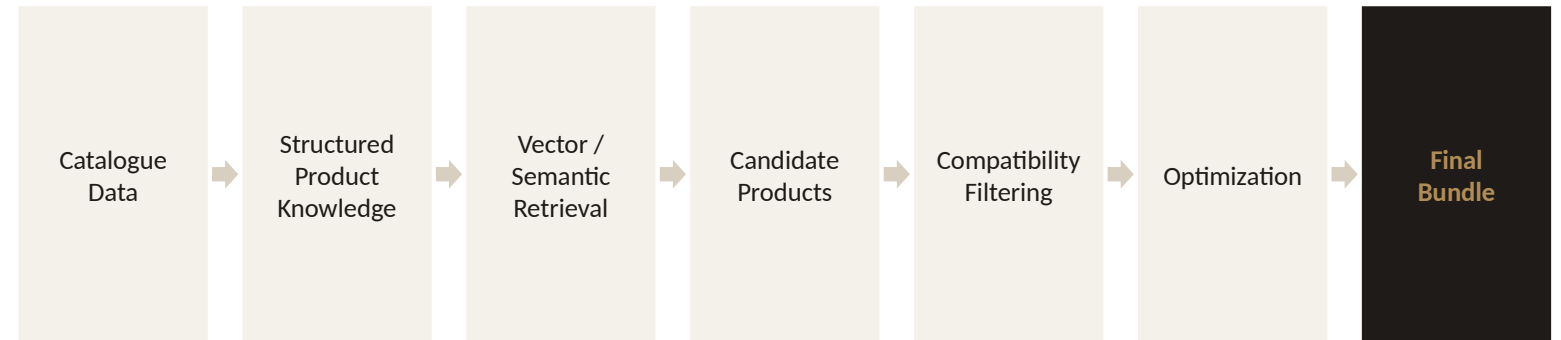
Grounding rule — the model is never allowed to invent product specifications — recommendations trace back to verified catalogue data.

Product Catalogue Intelligence

Attributes Retrieved

- Product category
- Dimensions
- Technical specifications
- Features
- Installation requirements
- Style / design attributes
- Price / budget where available
- Compatibility attributes

Retrieval → Decision Flow



The catalogue stays grounded in verified product information — retrieval identifies candidates before optimization ever runs.

Optimization Strategy

Objective — find a bathroom product bundle that maximizes overall suitability while satisfying user constraints.

Bundle Score = Compatibility + Preference Match + Design/Theme Match + Space Suitability + Feature Relevance – Budget Penalty – Constraint Violations

Hard Constraints — cannot be violated

- Product must fit available space
- Required product categories must be satisfied
- Budget must not exceed the specified limit
- Products must satisfy compatibility requirements

Soft Constraints — can be traded off

- Preferred aesthetic theme
- Feature preferences
- Brand / product preferences
- Design coherence
- Additional convenience features

Why AI Is Required

A conventional product filter cannot reason across every one of these variables at once:



As product categories and constraints grow, the number of possible combinations increases rapidly.

AI understands — unstructured inputs — natural-language requirements, bathroom images, design preferences

Retrieval grounds — recommendations in verified catalogue data

Optimization decides — the structured selection problem

AI Understanding + Verified Retrieval + Optimization — three complementary capabilities, combined.

Indian Market Context & Differentiation

KOHLER operates in a competitive Indian bathroom & sanitaryware market.

Competitive landscape

Jaquar	Hindware
Cera	Parryware
Grohe	Roca

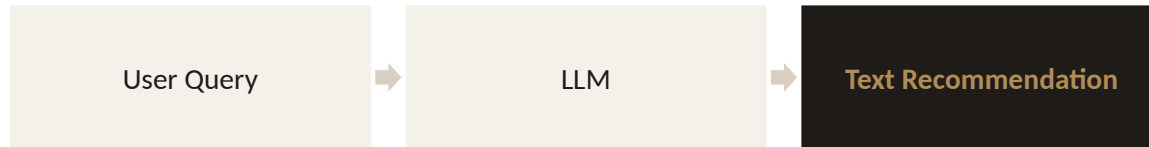
Context only — not a ranking or superiority claim. It establishes why an intelligent planning experience can differentiate.

Where this sits

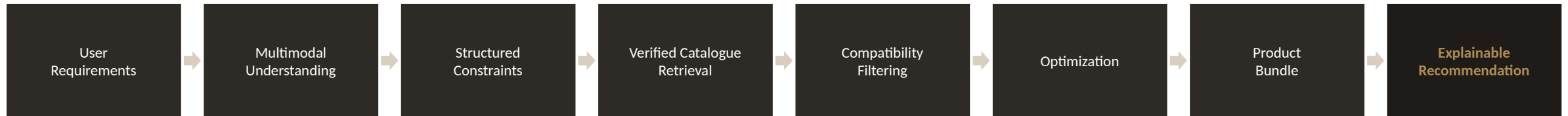
1. Traditional catalogue / product discovery
2. Product recommendation
3. AI-assisted bathroom planning
4. Constraint-aware complete-bundle recommendation

Technical Differentiation

Generic AI Chatbot



Proposed System

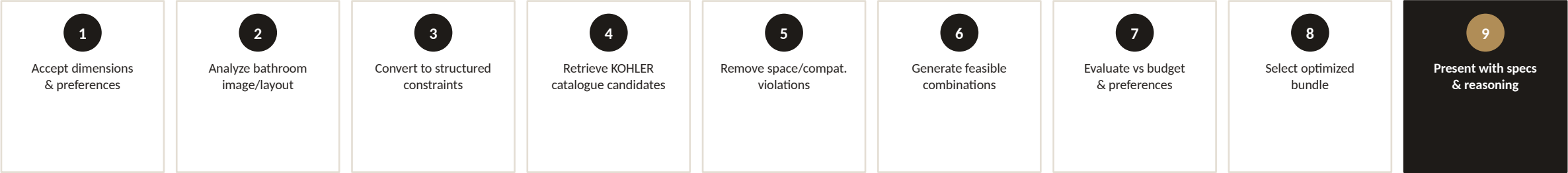


The proposed architecture combines generative AI with deterministic components rather than depending entirely on an LLM.

Example End-to-End Case

Bathroom size 8 ft × 6 ft	Budget ₹2,00,000	Theme Japanese Zen	Requirements Smart toilet, faucet, thermostatic shower, vanity
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System process



Demonstrates — *a complete configuration problem being solved — not isolated product recommendations.*

Feasibility & Business Value

Feasibility — modular by design

- Frontend for interaction
- Backend for orchestration
- Amazon Bedrock / Nova for AI
- Image processing for spatial understanding
- ChromaDB for retrieval
- Optimization engine for constraint satisfaction
- Structured catalogue data for product grounding

Business Value

- Reduces product-selection complexity
- Personalizes recommendations
- Helps customers understand suitable combinations
- Supports complete-bathroom configuration, not isolated discovery
- Connects requirements to relevant catalogue products
- Creates an AI-assisted digital planning experience

Note — each component has a defined, independent responsibility — no unsupported claims on revenue, conversion or market share.

Scalability & Prototype Demonstration

Scalability — beyond the prototype

- Larger product catalogues • additional bathroom categories
- More detailed spatial reasoning • installation constraints
- Regional availability • price/availability updates
- More sophisticated optimization objectives
- Personalization from customer history
- Integration with digital commerce / showroom workflows

Prototype — what will be shown

- Enter dimensions, upload image, set budget & theme
- System retrieves relevant catalogue products
- Candidates filtered for hard constraints
- Optimization generates a feasible bundle
- Recommended products displayed with reasoning

Design principle — *every component can be upgraded independently — the architecture doesn't need a redesign to grow.*

THE FULL NARRATIVE

Problem → Opportunity → Solution → Innovation → Architecture
→ Methodology → Market Context → Example → Prototype → Scalability

Problem

Multiple interacting constraints

Opportunity

AI simplifies & personalizes planning

Solution

KOHLER AI Bathroom Designer & Planner

Innovation

Multimodal AI + retrieval + compatibility + optimization

Architecture

Frontend + Backend + Bedrock/Nova + Vision + ChromaDB + Optimization

Methodology

Retrieve candidates → apply constraints → optimize bundle

Market Context

Competitive Indian sanitaryware market

Example

One complete end-to-end planning scenario

Prototype

Working user workflow

Scalability

Evolves into a broader AI planning platform

From product search to intelligent bathroom configuration.